

INTERNSHIP CAPSTONE PROJECT

# Hospital Management System

An Enterprise-Grade full-stack web application designed for high-performance scheduling, administrative analytics, and dynamic record transactions built with the MERN Ecosystem.

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 React + Vite | Node.js |  
MongoDB

 Intern  
Presentation

 MUI Design  
System



# Architectural Objectives

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## Centralized Registry

Architect robust and decoupled collections for healthcare records, linking doctors, patients, and scheduling registries with persistent document tracking and validation protocols.



## Real-Time Analytics

Implement operational dashboards rendering dynamic business intelligence, instant database count updates, and time-series weekly scheduling metrics using SVG integrations.



## E2E Optimization

Create a highly optimized client experience featuring accelerated application compiles, automated build trees, error-handling services, and standard QA pipelines.



# Production Tech Stack

| ARCHITECTURE LAYER   | CHOSEN TECHNOLOGY  | STRATEGIC RATIONALE & TECHNICAL ADVANTAGE                                                                    |
|----------------------|--------------------|--------------------------------------------------------------------------------------------------------------|
| Frontend Framework   | React 18 + Vite    | Extremely fast compiler optimization over classic bundlers; renders robust virtual DOM trees efficiently.    |
| UI Ecosystem         | Material UI (MUI)  | Provides dynamic, standardized grid elements, responsive typography models, and scalable admin themes.       |
| Analytics Engine     | Recharts           | A highly performant declarative chart framework processing responsive vector maps with native SVG rendering. |
| Backend Architecture | Node.js + Express  | Event-driven, asynchronous event handling, perfect for building high-concurrency stateless APIs.             |
| Database Engine      | MongoDB + Mongoose | Highly adaptable schema structures paired with an abstraction layer for model verification.                  |



# System Core Modules

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## 1. Operational Dashboard

Combines visual components and statistical monitors into an interactive interface. Aggregates system metrics (Total Patients, Active Specialists) and tracks weekly workflows with responsive SVG charting models.

## 2. Relational CRUD Registries

Decoupled interfaces managing profiles and consultation states. Features strict front-end state management using client routes to handle complex multi-step forms and persistent document creation safely.



# Decoupled REST API Integration

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- ✓ **Standardized Controller Routing:** Frontend calls access standard REST endpoints using an configured Axios client layer (e.g., `/api/doctors`, `/api/patients`, `/api/appointments`).
- ✓ **Verbs and Document Mutations:** Standardized operations using corresponding API calls (GET to retrieve arrays, POST to execute database creations, PUT for dynamic state changes, DELETE to drop rows).
- ✓ **Stateless Routing Protocols:** Separates server logic, keeping database connections scalable and reliable under heavy user requests.
- ✓ **Mongoose Middleware Schema Checks:** Validates database values on the server before completing insertions to prevent partial records and maintain consistent databases.



# QA & Compilation Workflow

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# System Transaction Performance

100%

API Transaction Completion Rate

## Zero-Latency State Synchronization

By leveraging an optimized asynchronous framework, the interface maintains seamless transitions and dynamic updates. Data payloads bypass heavy network loops via localized cache routing.

Search Query Time Complexity Formula:

$$T(n) = O(\log n)$$



# Weekly Appointment Load Trend

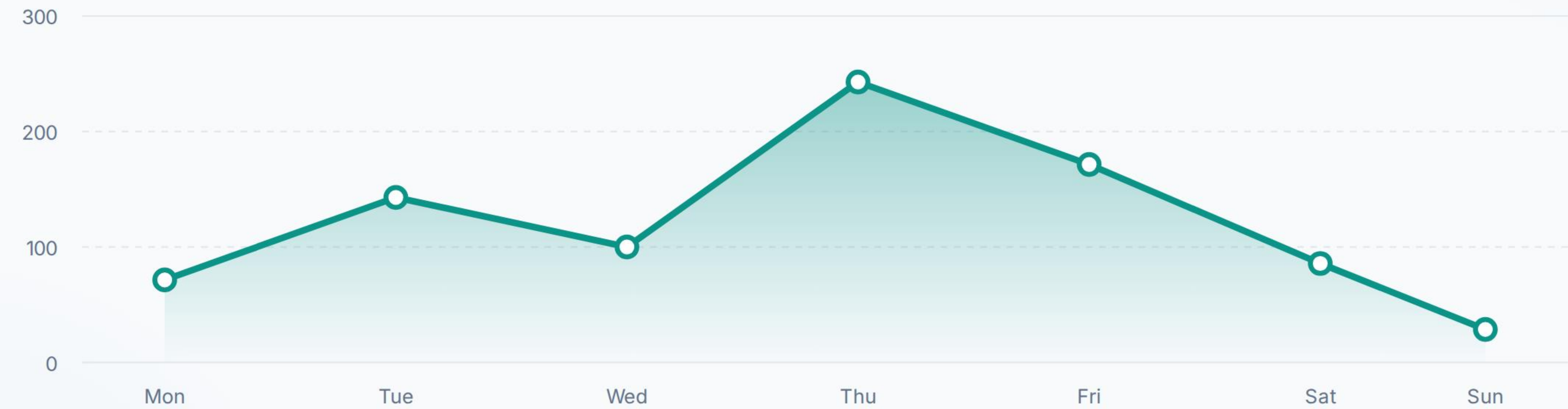
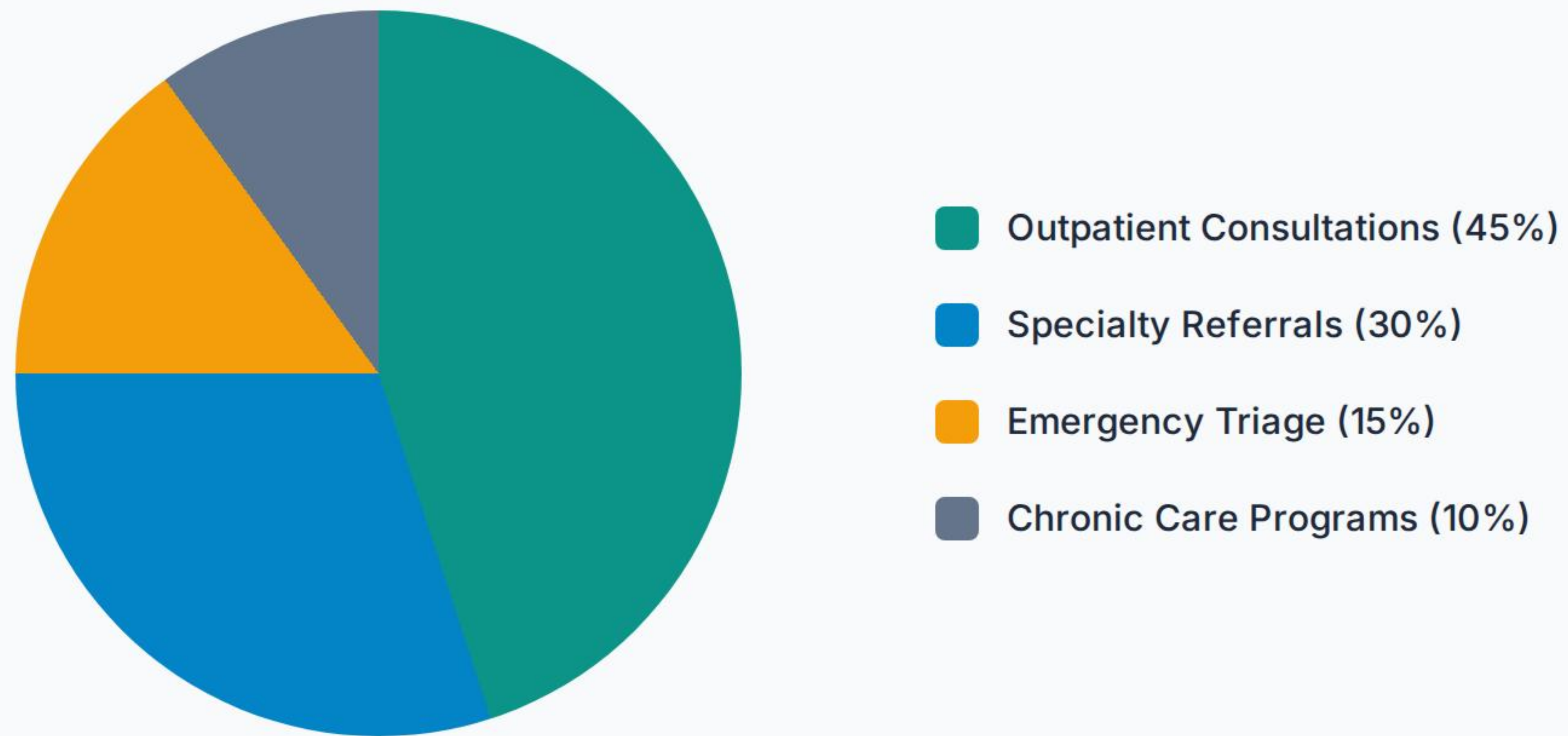


Figure 8.1: Weekly database scheduler transaction charts. System activity peaks mid-week on Thursday, tracking real-time operational capacity indexes.



# Active Patient Allocations

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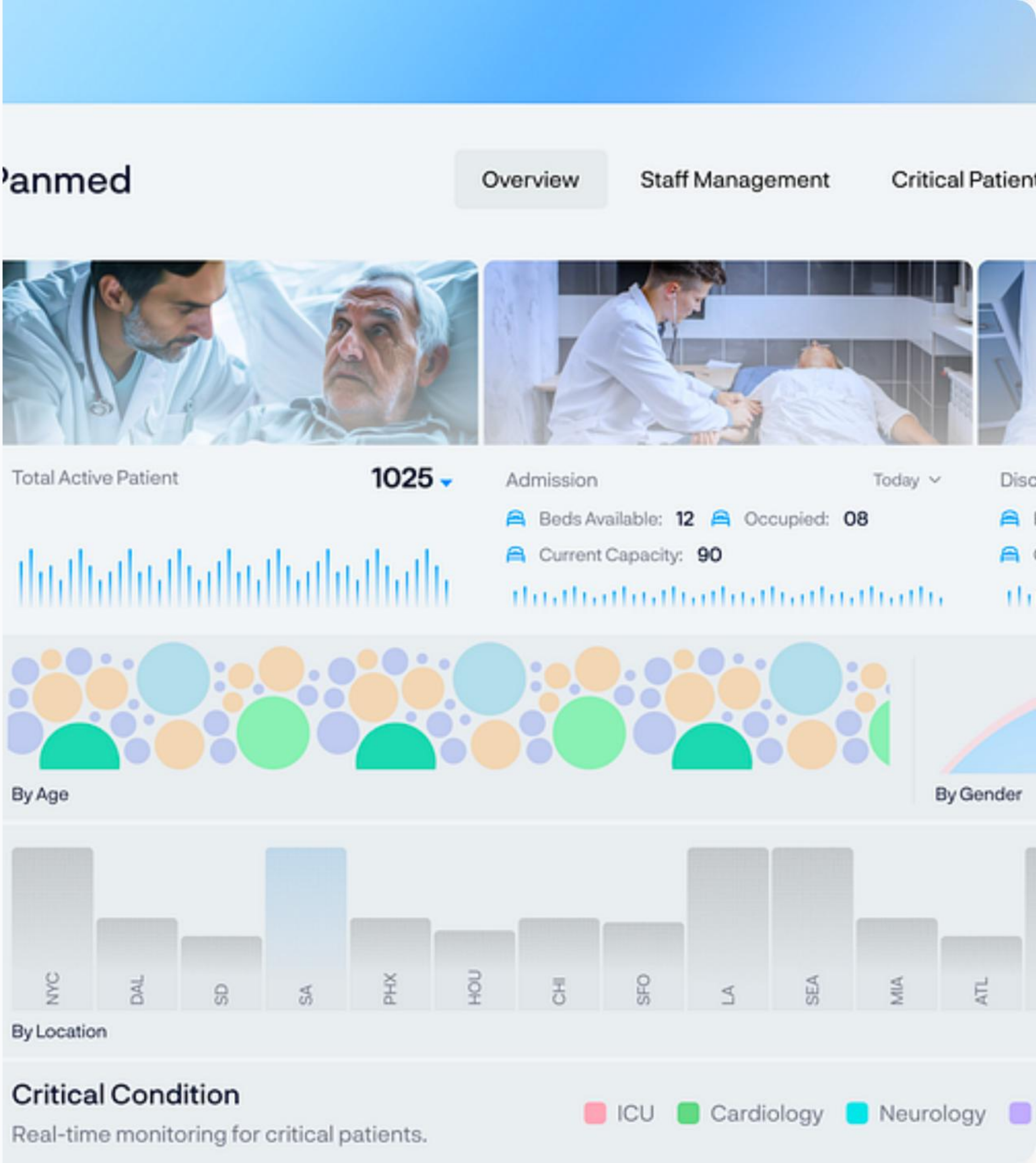




# Live Demonstration

The interactive dashboard application connects a lightweight state engine on the front end directly to our scalable MongoDB backend.

During the live session, we will verify instant operational metrics calculations, run through our strict error validation forms, and showcase our clean responsive admin design system.





# Engineering Roadmap

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## Secure Authentication

Integrate modern JWT (JSON Web Tokens) paired with dynamic Role-Based Access Control (RBAC) to enforce isolated routes between administrators, physicians, and patient users.



## Clinical Billing Node

Develop financial modules processing automated digital invoicing and medical coverage calculations, outputting clean downloadable invoices for patient records.



## Automated Reminders

Integrate cloud-hosted communication gateways (e.g., Twilio API) to send automated confirmation alerts, status changes, and follow-up notices to patient contact points.



# Questions & Discussion

Thank you for your valuable time and consideration. Open for feedback and architecture reviews.



## Email Contact

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## Repository Workspace

[github.com/intern/hms-mern](https://github.com/intern/hms-mern)



## Platform Status

HMS Sandbox v1.0.0 (Stable)



# Image Sources

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<https://cdn.dribbble.com/userupload/46616377/file/1fc785129cb7b226b2c3171f53e5cf55.png?resize=752x&vertical=center>

Source: [dribbble.com](https://dribbble.com)